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Tectonic rotations and translations in western Canada; new evidence from Jurassic rocks of Vancouver Island*

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Summary. The Lower Jurassic Bonanza Group (Sinemurian and Pliensbachian lavas and tuffs about 205 to 195 Ma) on northwestern Vancouver Island possesses a stable remanent magnetization with a mean direction (**D**, **I**) of 276°, 42° (13 sites distributed through 1800 m of section, 128 specimens, $\mathbf{k}_B = 50$, $\alpha_{95} = 6°$) palaeopole 22° N, 154° E (*dm*, *dp* = 9°, 6°). Reversals are present, but one polarity is dominant and most directions are westerly. The palaeolatitude is 24 ± 6°, with the usual ambiguity as to whether it is north or south of the palaeoequator. To determine the relative motions between ancestral North America and Vancouver Island, these data are compared with determinations from peri-Atlantic igneous rocks dated radiometrically in the same time interval. The use of model apparent polar wander paths for this purpose is criticized. A net, post-Jurassic northerly motion of 9 ± 7° or 57 ± 7° is indicated, depending on the chosen hemisphere of origin. The net post-Jurassic rotations, relative to cratonic North America, are 66 ± 7° anti-clockwise or 114 ± 7° clockwise for the northern and southern hemispheres, respectively. Such large rotations are, apparently, a common feature of early Mesozoic rocks of the Canadian Cordillera. It is suggested that these large and variable rotations characterize the deformation occurring during the pre-mid-Cretaceous amalgamation of the various terranes of British Columbia.

Key words: Tectonic rotations, palaeomagnetism, Vancouver Island

1 Introduction

Post-Late Triassic displacement of Vancouver Island from the south with respect to ancestral North America has been postulated, based on anomalously low palaeolatitude (Irving & Yole 1972; Yole & Irving 1980; Schwartz, Muller & Clark 1980) and the occurrence of disjunct marine faunas (Tozer 1970, 1982). Anomalously low palaeolatitude also has been determined from Upper Triassic rocks of southern Alaska. This, and geological similarities

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with Vancouver Island and the Queen Charlotte Islands have led to the idea that these regions, now spread over about 1500 km, were formerly close together and formed a compact terrane – Wrangellia – which was situated to the south of their present positions relative to North America (Jones, Irwin & Ovenshine 1972; Jones, Siberling & Hillhouse 1977; Hillhouse 1977; Stone, Panuska & Packer 1982; Hillhouse & Grommé 1984). Because reversals of the geomagnetic field were frequent in the Late Triassic, it is not possible to determine the hemisphere or origin, so the net displacement since the Late Triassic could have been something in excess of 1000 km from northern palaeolatitudes or 5000 km from southern palaeolatitudes.

Determinations of relative displacement critically depend on the accuracy with which the reference palaeopoles of the craton are known. Recent revisions of the geological time-scale (Harland *et al.* 1982; Kent & Gradstein 1985) have transferred what used to be the main source of Late Triassic reference information (the Newark Supergroup) from the Late Triassic into the Early Jurassic. Moreover, palaeopoles from older (Early Jurassic) and from younger (Middle Jurassic) Newark rocks are significantly different, showing that the earlier Jurassic was an interval of rapid apparent polar movement (Smith & Noltimier 1979). The consequence of these revisions is that during the Late Triassic and Early Jurassic, cratonic North America may have been situated at palaeolatitudes 5 or 10° lower than those previously assigned. Hence relative displacements of parts of the Canadian Cordillera,

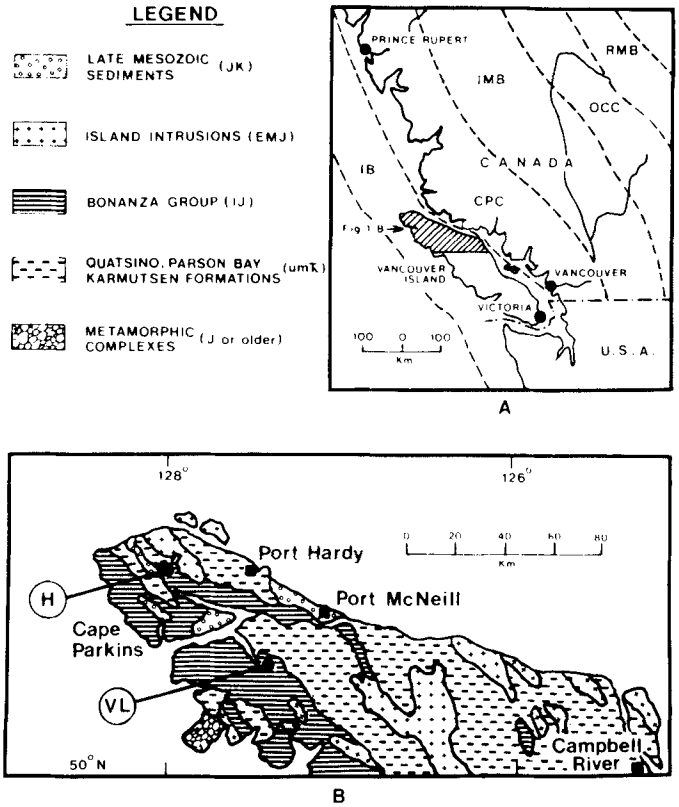


Figure 1. Index map showing position of Vancouver Island and western cordillera tectonic belts, IB Insular Belt, CPC Coast Plutonic Belt, IMB Intermontane Belt, OCC Omineca Belt, RMB Rocky Mountain Belt, H Holberg, VL Victoria Lake.

determined from early Mesozoic data by Yole & Irving (1980) and Monger & Irving (1980), may have been less than they estimated. Using this and other lines of evidence, Gordon, Cox & O'Hare (1984) have questioned the presence of such displacements at all. The purpose of this paper is to present new data and new analyses relevant to this discussion. It will be contended that for periods of rapid apparent polar motion, such as the Jurassic, data used for comparison should be restricted to intervals not longer than approximately one or two geological stages, say 10 or 15 Ma. As we shall see, this ideal procedure cannot at present be fully realized although a reasonable approximation to it can be made.

2 Geology and sampling

The Lower Jurassic Bonanza Group overlies the Upper Triassic Vancouver Group. It is overlain in northern Vancouver Island by the Lower Cretaceous Longarm Formation or by the Queen Charlotte Group. The Bonanza Group is an arc sequence, comprising calc-alkaline lavas, tuffs and breccias of basaltic to rhyo-dacitic composition (Gunning 1932; Hoadley 1953; Jeletzky 1976; Muller, Northcote & Carlisle 1974; Muller 1977, 1983). Also present are interbedded argillites and greywackes containing Early Jurassic marine fossils of Sinemurian and Pleinsbachian age (Jeletsky 1970, 1973, 1976; Ferbold & Tipper 1970). This corresponds to an interval of 194 to 206 Ma on the time-scale of Harland *et al.* (1982), and 193–204 Ma on the time-scale of Kent & Gradstein (1985). K/Ar whole-rock dates range from 103 to 161 Ma (Muller *et al.* 1974), much younger than the ages determined from fossils, indicating that one or more thermal events have caused argon loss. This is to be expected in a region that has had such a complex geological history. Hence these K/Ar data cannot be regarded as reliable indicators of age.

The Bonanza Group outcrops widely on Vancouver Island (Fig. 1). Of the many localities visited in 1980, 1981 and 1983, the Cape Parkins shore-line section (Muller *et al.* 1974) was selected for detailed study because of the thick section, good exposure, relatively simple structure, and reliable fossil age-determinations. Moreover, the section contains abundant lavas and hypabyssal intrusions which might be expected to record accurately the geomagnetic field at the time of cooling. Several road-cuts and quarries in the Holberg Inlet and Victoria Lake areas also were sampled, but inland it was not possible to sample thick continuous sequences. The rocks have undergone low-grade regional metamorphism, a maximum of zeolite facies according to Muller *et al.* (1974).

The Bonanza Group is intruded by the granitoid Island Intrusions (Northcote & Muller 1972). Minimum ages for the intrusions are fixed by overlying sedimentary rocks ranging from Callovian (Jeletzky 1976) to Valanginian (Muller *et al.* 1974). The intrusions have yielded K/Ar ages ranging from 148 to 181 Ma, overlapping the K/Ar ages for Bonanza volcanics (Muller *et al.* 1974). Pb/Pb dates obtained from the intrusives range between 180 and 190 Ma (Isachsen *et al.* 1985), indicating an age of latest Early Jurassic or earliest Middle Jurassic. Evidently the Island Intrusions are contemporaneous with, or somewhat younger than, the Bonanza Group. Smaller, Porphyritic intrusions, possible feeders for some of the Bonanza volcanics, cut stratified rocks of the Bonanza Group. The probable minimum date for cooling, uplift and unroofing of the Island Intrusions is mid-Cretaceous, as indicated by the presence of abundant granitoid clasts in conglomerates of this age. No granitoid pluton is known to occur close to the sampling sites, but there are small bodies about 5 km from the northernmost Victoria Lake site across a fault. There was Tertiary igneous activity in the region. An Early Tertiary quartz diorite body occurs near Zeballos, 70 km SE of the sampling area. There also are small outcrops of Late Tertiary basalts betwen Victoria Lake

and Port McNeil, the closest being 10 km NE of the northern Victoria Lake sites. Vertical Tertiary dykes occur sporadically.

This history indicates that there are several possible events that could have affected the magnetization of Bonanza rocks since they were deposited. The most intense effects would be expected from the Island Intrusions. Also, an overprint of Cretaceous or even Tertiary age, acquired when these intrusions and their host rocks were uplifted, may be anticipated. Whether the small porphyritic bodies which cut the Bonanza represent an intrusive episode separate from the Island Intrusion is uncertain, but they too could have caused re-heating not long after deposition of the Bonanza. Finally, there may have been thermal affects related to Tertiary activity. However, a Tertiary dyke sampled at site 9 to test for such a possibility has remanence directions very different from those commonly observed in Bonanza rocks (Table 2) showing that overprinting by such activity was negligible.

Sampling sites were located in fault-bounded blocks or 'panels' with variable attitudes. The beds are tilted but not folded, so corrections for plunge are not needed. The main faults trend northwesterly. They are steeply dipping and affect rocks as young as Late Cretaceous, and are probably latest Cretaceous or Early Tertiary (Muller *et al.* 1974). Less common, northeasterly trending faults are younger, because in places they offset the northwesterly faults and appear to have some spatial relationship to Late Tertiary volcanism.

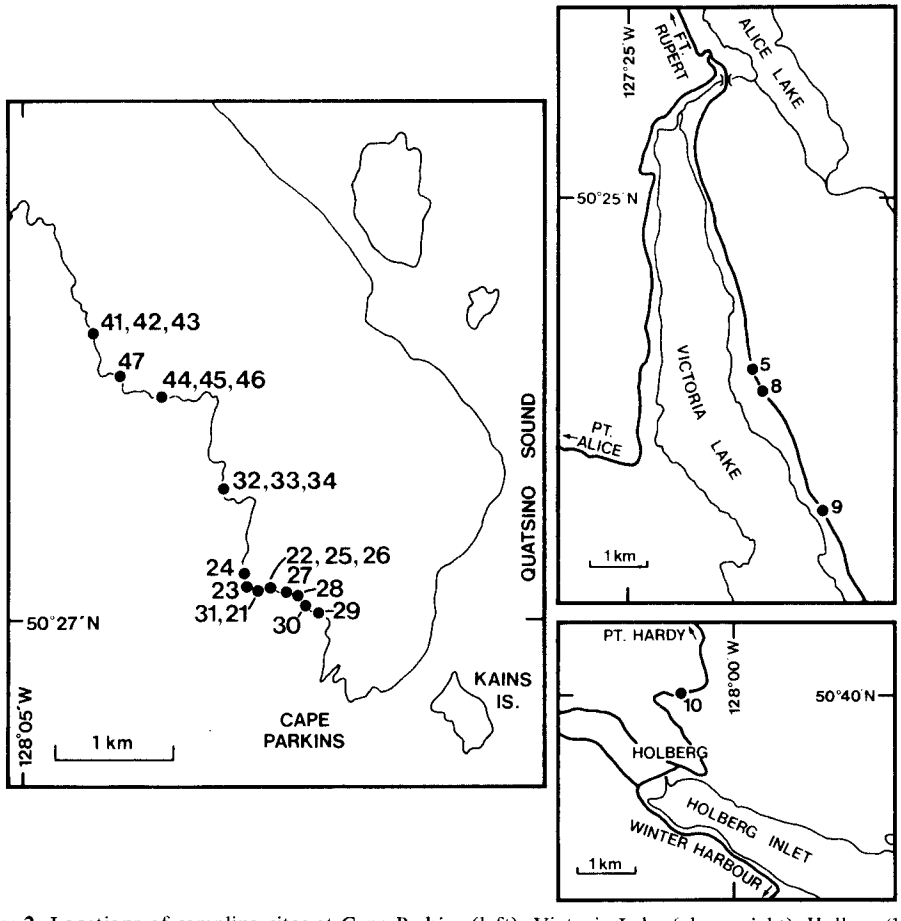


Figure 2. Locations of sampling sites at Cape Parkins (left), Victoria Lake (above right), Holberg (below right).

Estimates of palaeohorizontal were obtained by making several determinations of bedding at each sampling site by measuring the attitude of interbedded sedimentary rocks or flow surfaces. Except for sites in or close to well-stratified sedimentary rocks, 'local' attitudes obtained in this way may contain errors caused by, for example, appreciable original slopes in lavas. Such local errors were minimized by calculating mean dips for each block or panel. The better grouping of magnetic directions thus derived warrants this approach (Tables 1 and 2).

Sampling sites in the Cape Parkins section (Fig. 2) extend over an estimated stratigraphic thickness of about 2100 m, assuming no significant repetitions by faulting. Each site is an individual lava flow, a sedimentary bed, or dyke. The sites are all above unit 2 and below unit 10 of Muller *et al.* (1974). Dips are to the WNW. Dips in the upper 1800 m range from 40° to 60°, those in the lower 300 m from 65° to 85°. Rock types sampled include thinly-bedded reddish tuffs, dark andesitic basaltic flows and breccias, and fine to medium-grained basic to siliceous dykes. Orientations were done by sun and magnetic compasses, the two agreeing to 2° or better. About 5 cores were taken at each site and generally two specimens were cut from each core as detailed in Table 2.

Table 1. Sampling sites

Site	Rock-type	$BD^{\circ}_{L}/BBDA^{\circ}_{L}$	$BD^{\circ}_{P}/BBDA^{\circ}_{P}$	UTM grid ref.
43	red tuff	50/305(1)	48/301	655914
42	red tuff	45/302(1)	48/301	655914
41	red tuff-breccia	40/295(1)	48/301	656913
47	very fine-grained lava	40/282(2)	40/282	658909
44	coarse feldspar porphyry	60/310(2)	53/301	662907
45	coarse feldspar porphyry	60/310(2)	53/301	662907
46PO	coarse feldspar porphyry	47/290(1)	53/301	662908
46LA	fine-grained lava	47/290(1)	53/301	662908
32	amygd. haematitic lava	60/290(3)	57/292	667899
33	fine-grained lava lense	55/295(3)	57/292	667899
34	basalt dyke	55/295(3)	57/292	667899
24	feldspar porphyry dyke	45/282(2)	50/282	668893
23	amygd. haematitic lava	60/282(?)	50/282	669892
31	amydg. haematitic lava	60/275(2)	50/282	670892
21	f.g. haematitic lava	50/290(2)	50/282	670892
22	fine-grained dyke	50/290(2)	50/282	671892
25	fine-grained lava	45/282(2)	50/282	671892
26	red scoriaceous lava	45/282(2)	50/282	671892
27	amygdaloidal lava	85/310(1)	77/293	672891
28	green and red tuff	85/310(1)	77/293	673891
30	red tuff	80/290(1)	77/293	674899
29	amygdaloidal lava	65/280(1)	77/293	675899
05	amygdaloidal lava	22/170(3)	28/220	146831
08	porphyritic lava	17/170(3)	28/220	148827
10	medium-grained lava	17/205(3)	17/205	699131
09	f.g. siliceous dyke	–	–	159807

Notes: Site numbers have no stratigraphic significance. Sites from Cape Parkins (Fig. 2) are listed in stratigraphic order from 43 (youngest) to 29 (oldest). Sites 05, 08, 09 are from Victoria Lake, and 10 from Holberg (Fig. 2). $BD^{\circ}_{L}/BBDA^{\circ}_{L}$ are bedding dip and direction of dip as determined at each site. The numbers in brackets indicate the relative reliability of bedding attitudes, (1) good, (2) fair (3) poor. $BD^{\circ}_{P}/BBDA^{\circ}_{P}$ are bedding attitudes averaged over a rock-panel within which the measured attitudes are approximately uniform. 'Panel' attitudes for sites 05 and 08 are from Muller *et al.* (1974). UTM grid references have the prefix 9UWF, except those for sites 05, 08 and 09 which have the prefix 9UXF, and that for site 10 which has the prefix 9UWG.

3 Laboratory procedures

At some collecting sites the initial (nrm) directions of magnetization were well-grouped. Thermal and alternating field (af) demagnetization of pilot specimens showed that, after removal of a small magnetization directed close to the present Earth's field (PEF), only a single magnetization is present. Orthogonal plots show demagnetization trajectories going to the origin. The remaining specimens from these sites were demagnetized at four steps, sufficient to confirm that only single magnetizations are present. Data were then selected at points centrally placed along linear trends, in the range 10–80 mT or 300°–600°C.

At other sites, the nrm directions varied from being poorly to well-grouped, and studies of pilot specimens showed that, in addition to small PEF magnetizations removed during the early stages of demagnetization, two magnetizations are present in each specimen. One magnetization has unblocking temperatures in the range 500°–600°C, typical for magnetite, and the other is characterized by unblocking temperatures of 600°–700°C, indicative of haematite. All specimens at these sites with complex magnetization were demagnetized in many steps (usually 12 or more) in order to isolate the magnetizations present. The lower unblocking temperature magnetization (generally carried by magnetite) was determined by calculating the magnetization removed along linear segments of orthogonal plots (Zijderveld 1967). After this procedure, a more resistant magnetization (generally carried by haematite) can sometimes be recognized as a crudely defined stable end-point, but its magnitude is usually so small that its direction cannot be determined accurately. Although, in addition to PEF magnetization, two other magnetizations are commonly present, it is seldom possible to separate both accurately, only the principle magnetization is extractable.

4 Cape Parkins section

Three magnetizations were identified. The first (**A**) is directed downwards (relative to bedding) toward the west or west northwest. The second magnetization (**B**) is directed

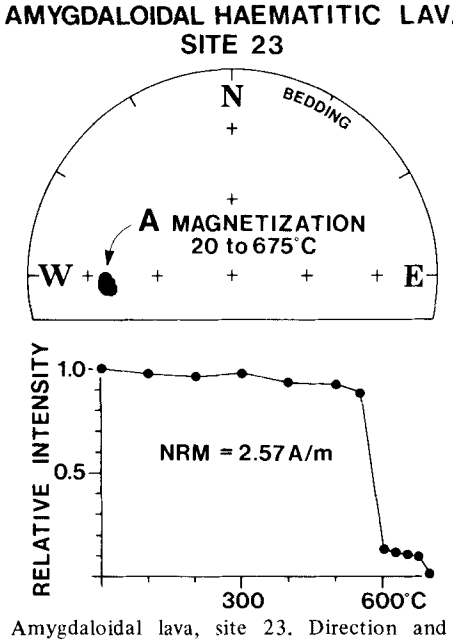


Figure 3. A magnetization. Amygdaloidal lava, site 23. Direction and intensities during thermal demagnetization. Solid circles indicate downward inclination. Directions constant throughout.

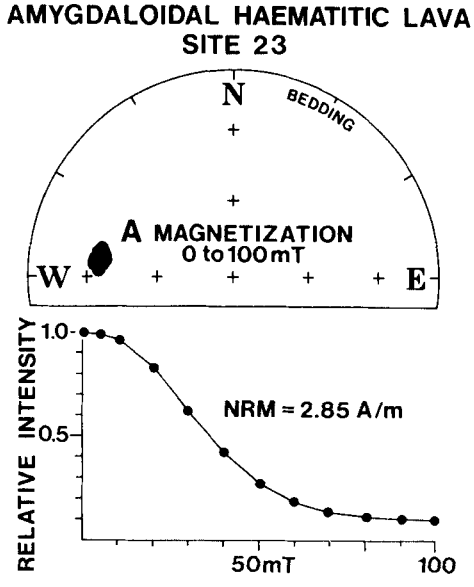


Figure 4. A magnetization. Amygdaloidal lava, site 23. Direction and intensities during af demagnetization. Inclination downward. Directional stability with coercive forces mainly between 20 and 70 mT.

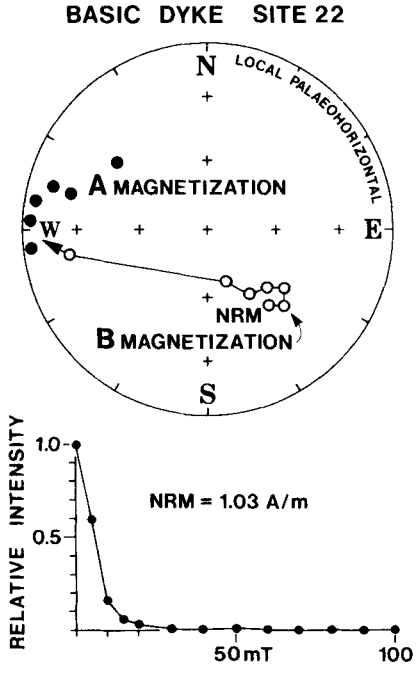


Figure 5 B magnetization with subordinate A magnetization (poorly defined) in basic dyke, site 22. Large direction and intensity changes occur during af demagnetization. Solid (open) circles denote downward (upward) inclination.

upward, roughly antiparallel to **A**. The third (**C**) points to the northwest and is close to the plane of bedding. **A** occurs throughout the section and is by far the most common. **B** dominates at one site (22), but elsewhere occurs only sporadically. **C** is confined to the more deformed, steeply dipping beds in the lower part of the section.

A magnetization at site 23 is shown in Figs 3 and 4. The direction is unchanged during af and thermal demagnetization up to 100 mT and 675°C. Most of the magnetization has unblocking temperatures between 500 and 600°C and remanent coercive forces in the range 20–70 mT, typical for low-titanium magnetite. Polished sections show abundant fine-grained magnetite. About 10 per cent of the magnetization has unblocking temperatures between 675 and 700°C, and remanent coercive forces in excess of 100 mT, and is presumably carried by haematite. Haematite occurs in polished section as specularite grains,

Table 2. Results by sites

Site	$C(S)$	M (A m^{-1})	D°_L, I°_L	D°_p, I°_p	k_w	α°_{95}	Type
43	5(9)	0.32	285, 46	280, 47	80	6	**A**
42	5(8)	0.17	282, 50	282, 47	239	4	**A**
41	9(16)	0.16	283, 44	290, 37	139	3	**A**
44	4(7)	0.94	283, 27	274, 30	32	11	**A(B)**
45	7(12)	0.67	299, 45	285, 50	18	11	**A(B)**
46PO	3(06)	1.35	253, 48	268, 48	143	6	**A(B)**
46LA	3(5)	0.94	241, 27	252, 30	185	6	**A**
32	5(8)	0.66	285, 50	288, 54	152	5	**A**
33	7(13)	0.71	284, 51	280, 49	55	7	**A**
34	5(9)	0.44	276, 45	273, 42	6	23	**A(B)**
24	4(7)	0.08	312, 31	311, 27	12	18	**A**
23	7(10)	2.73	269, 37	270, 32	191	4	**A**
31	6(10)	1.62	275, 28	282, 38	286	3	**A**
21	5(6)	0.11	272, 40	264, 38	173	5	**A**
22	6(11)	1.14	136, −48	127, −51	119	4	**B(A)**
25	5(8)	0.37	282, 47	282, 42	22	12	**A(B)**
26	4(4)	0.62	260, 57	263, 52	17	23	**A**
28	3(3)	0.07	318, 16	298, 26	30	23	**C**
30A	1(1)	0.73	285, 45	290, 48	–	–	**A**
30C	3(5)	0.01	320, −6	322, −5	51	11	**C**
29	5(8)	0.32	298, 17	308, 02	62	7	**C**
05	3(5)	1.89	29, 30	20, 39	48	11	**A**
08	3(6)	2.25	37, 27	30, 43	126	6	**A**
10	5(10)	0.51	359, −49	–	34	8	**B**
09	6(8)	1.05	148, −66	–	67	7	–

(1) $C(S)$ are the number of cores and specimens. M is the intensity of magnetization. D°_L, I°_L, is the mean site direction calculated giving specimens unit weight and relative to the 'local' bedding given in Table 1. D°_p, I°_p, is the same relative to the 'panel' bedding given in Table 1. k_w and α°_{95} are estimates of precision and the radius of the circle of confidence (Fisher 1953). The type of magnetization, as described in the text, is given in the last column; (2) Experimental procedures are used as follows: site 43, all specimens cleaned at 600°C; 42, 600°C; 41, 600°C (15 specimens), 80 mT (1 specimen); 44, 550°C; 45, 550°C; 46 P, 550°C; 46 L, 550°C; 32, 650°C (1), 50 mT (7); 33, magnetization after treatment at 60 mT subtracted from that at 500°C (13); 34, 10–40 mT (6), 80 mT (2), 625°C (1); 24, 40 mT (6), 500°C (1); 23, 50 mT (9), 650°C (1); 31, 50 mT (9), 650°C (1); 21, 40 mT (5), 300°C (1); 22, 10–30 mT (10), 200–600°C (1); 25, 300–625°C (2), 40 mT (6); 26, 50–80 mT (3), 500–600°C (1); 28, 20–40 mT (2), 500–650°C (1); 30A, 80 mT; 30°C treatment at 100 mT followed by treatment at 600°C; 29, 500–675°C (1), 30–80 mT (7); 5, 50 mT (4) 500°C (1); 8, 80 mT (5) 500°C (1); 10, 75 mT (6) 50–70 mT (2) 500–675°C (2); 9, 10 mT. No useful data were obtained from sites 27 and 47 because it proved impossible to separate the magnetizations present.

and as rims in highly-altered reddened pyroxene. Both magnetite and haematite have identical directions of magnetization. Specimens with **A** magnetization were cleaned in fields 20–80 mT or 500–650°C as set out in the footnote to Table 2.

B magnetization is most readily observed in the fine-grained dyke at site 22. During af demagnetization the directions first cluster to the SE with upward inclination (Fig. 5). Then, at higher fields, they migrated westward toward the **A** direction, but with much scatter because the intensity fell by two orders of magnitude. **A** cannot be precisely determined, noise prohibiting the detection of an accurate endpoint. **B**, the principle magnetization of this dyke can be determined directly because **A** is so small. Companion specimens show similar behaviour during thermal demagnetization. Magnetite occurs in polished section as abundant blocky grains. Haematite was not identified, but the magnetic behaviour (Fig. 5) indicates that it is present in trace amounts. Evidently both magnetite (**B**) and haematite (**A**) contribute to the magnetization, but they have directions that are approximately antiparallel, not parallel as at site 23. Behaviour which is a mirror-image of that just described also is common. In these instances an **A** magnetization carried by magnetite predominates, and a very small oppositely-aligned **B** magnetization, presumably carried by haematite, is revealed at high temperatures and high fields (see right hand column of Table 2). The directions of the subordinate magnetizations (whether **A** or **B**) cannot be accurately determined. The principal magnetization can be obtained either directly, or, by subtracting the subordinate magnetization, that is, by calculating the magnetization removed along linear trends, the latter being used where direct cleaning yields errors in excess of one degree (footnote of Table 2).

The **C** magnetization characterizes the southern, more steeply tilted, lower part of the section where the beds are nearly vertical. It is always associated with red haematitic volcanigenic sedimentary rocks. Demagnetization up to 100 mT reduces the intensity by only 30 per cent, and subsequent thermal demagnetization up to 675°C produces a steady decrease in intensity. Throughout both demagnetization processes the directions are essentially unchanged, and thermal demagnetization yields linear trends towards the origin (Fig. 6). The magnetization is carried by haematite, which may be seen as abundant small

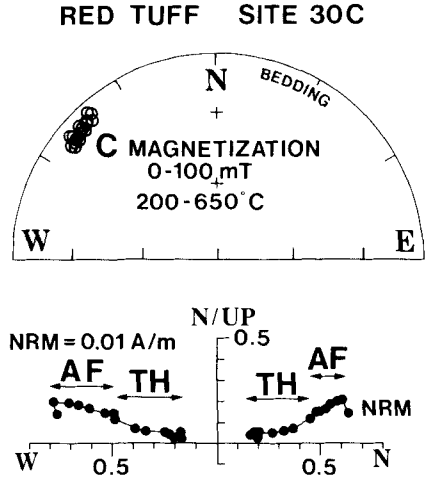


Figure 6. C magnetization. Weakly magnetized tuff, site 30. Directions (above) and orthogonal plots (below) of af demagnetization up to 100 mT followed by thermal demagnetization from 200 to 650°C. Minor directional changes during demagnetization.

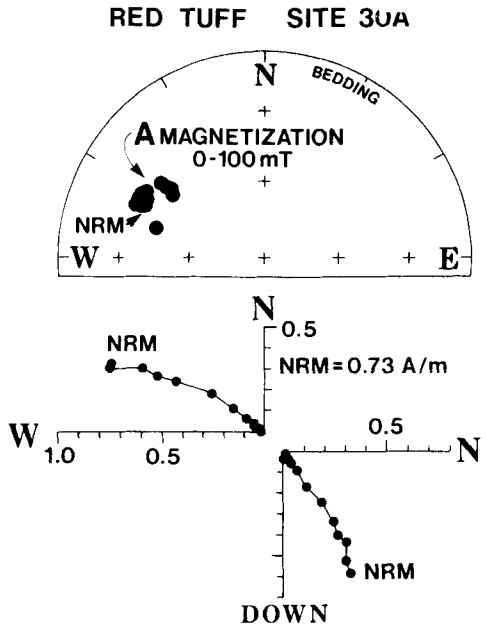


Figure 7. A magnetization. Strongly magnetized tuff, site 30. Directions above, orthogonal plots below. Points are at 10 mT intervals.

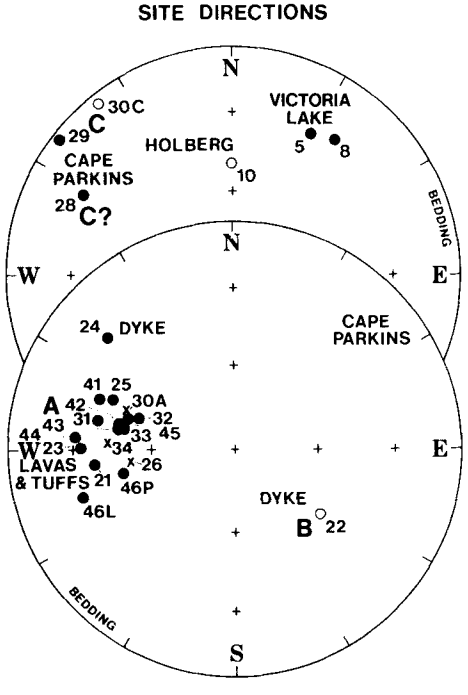


Figure 8. Site mean directions with reference to panel bedding. A and B magnetizations from Cape Parkins below. C magnetization from Cape Parkins and results from other locations above.

grains in polished section. One specimen of red tuff at site 30 has a much stronger magnetization, and possesses typical **A** magnetization with linear decay to the origin above 50 mT and unblocking temperatures of 500 -600°C (Fig. 7).

The directions of the three magnetizations observed in the Cape Parkins section are plotted in Fig. 8. In Table 2, directions are given with respect both to local and panel attitudes. Within-site precision ($\mathbf{k}_w$) is variable, and where it is low and the number of determinations few, the errors (α_{95}) in site-means sometimes exceed 20°. Generally, however, errors in site-means are between 5° and 10°. Data from lavas and red tuffs are well-grouped, and all are predominantly of one polarity, corresponding to the **A** magnetization. The porphyry dyke (site 24) has a magnetization with a similar direction to that in lavas and tuffs (**A**). The principal magnetization of the fine-grained dyke (site 22) is of opposite polarity (**B**). Although dyke directions are broadly comformable to those in the stratified sequence, declinations are perceptibly clockwise of the main group, indicating intrusion at a time, sufficient later to record either a change in the geomagnetic field direction or penecontemporaneous rotation.

Grand averages are given in Table 3, all calculated by giving unit weight to each site-mean direction. In entries 1 and 2, all sites with **A** or **B** magnetizations from Cape Parkins are included and averaged regardless of sign. In entries 3 and 4, poorly defined site directions are excluded (sites 34 and 26, with errors exceeding 20°, and site 30A (Fig. 7) at which **A** magnetization was found in only one specimen). In entries 5 and 6, data from dyke rocks for which attitudinal corrections may not be accurate (sites 24 and 22), also are excluded. Directions relative to panel bedding are always better grouped than when corrected to local bedding, indicating that the former are better estimates of palaeohorizontal. This is particularly evident in entry 6, in which those sites that are either magnetically or attitudinally poorly defined are excluded. Hence, entry 6 is regarded as the best summary of the Cape Parkins data. The stratigraphic thickness represented by entry 6 is 1800 m, which is less than the total thickness sampled (2100 m) because **C** magnetizations from the highly tilted lowermost beds (300 m) are excluded.

Table 3. Summary of Cape Parkins results

	Type	Ref.	$B(S)$	D°_M, I°_M	$\mathbf{k}_B$	α°_{95}	palaeopole lat.°, long.°	dm°, dp°
1. Lavas, tuffs and dykes, all sites	**A + B**	L	18(150)	280, 44	25	7	26, 152	9, 6
2. Lavas, tuffs and dykes, all sites	**A + B**	P	18(150)	280, 43	34	6	25, 152	7, 5
3. Lavas, tuffs and dykes, excluding those with $\alpha_{95} > 20°$	**A + B**	L	15(136)	281, 43	23	8	26, 151	9, 6
4. Lavas, tuffs and dykes, excluding those with $\alpha_{95} > 20°$	**A + B**	P	15(136)	281, 43	29	7	26, 150	10, 6
5. Lavas and tuffs excluding those with $\alpha_{95} > 20°$	**A**	L	13(118)	276, 43	31	8	22, 154	9, 6
6. *Lavas and tuffs, excluding those with $\alpha_{95} > 20°$*	**A**	*P*	*13(118)*	*276, 42*	*50*	*6*	*22, 154*	*9, 6*
7. Lavas and tuffs (sites 28 to 30)	**C**	L	3(016)	306, 10	9	34	26, 116	43, 22
8. Lavas and tuffs (sites 28 to 30)	**C**	P	3(016)	310, 08	16	31	27, 112	32, 16

Type of magnetization is given in the second column, followed by the reference plane used, whether it is that locally determined (L) or the average panel bedding (P). $B(S)$ are the number of collecting sites and specimens. D°_M, I°_M are the declination and inclination of means. $\mathbf{k}_B$ and α_{95} are Fisher's estimate of precision and radius of circle of confidence ($P = 0.05$). The latitude and longitude of the palaeopole and the semi-axes of the error ellipse (dm°, dp°) are given in the right-hand columns. Entry 6, in italics, is regarded as the best estimate of the mean direction of the middle Early Jurassic palaeofield.

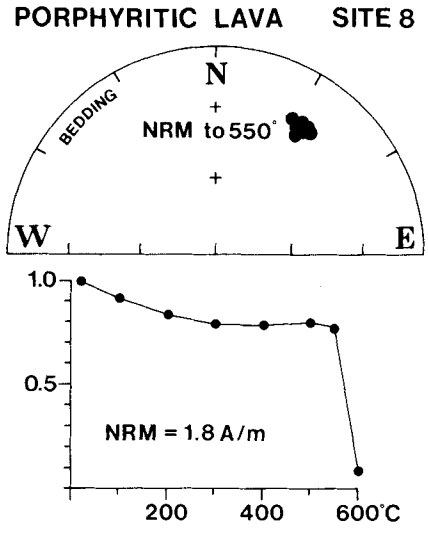


Figure 9. A magnetization. Porphyritic lava, site 8, Victoria Lake. Directions (above) and intensity (below) during thermal demagnetization. Inclination downward.

5 Inland localities

Lava (site 8) at Victoria Lake yielded a single component magnetization directed to the northeast with an inclination similar to that at Cape Parkins, but with a very different northeasterly declination (Fig. 9). Lava from nearby site 5 give similar directions (Table 2).

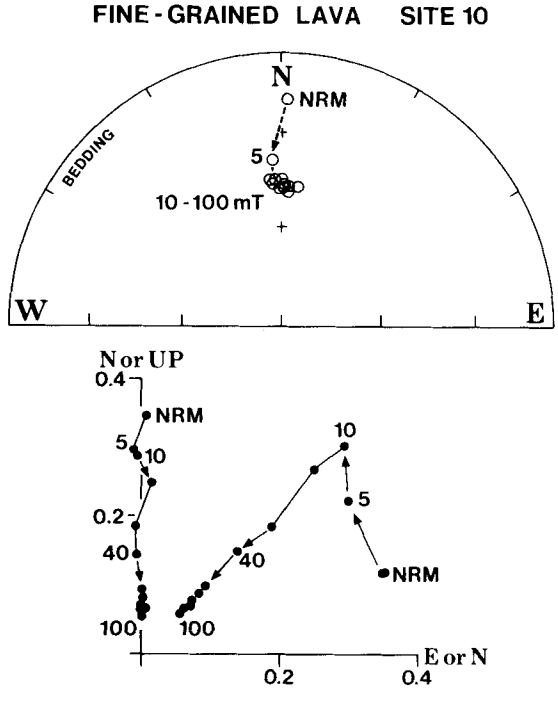


Figure 10. B magnetization. Lava, site 10, near Holberg. Direction changes above. Inclination upward. Orthogonal plots below N/E (on the left) and N/Upward (on the right).

Sites 5 and 8 are from the same stratigraphically continuous section. Lava from site 10 near Holberg provided excellent end-points after removal of a small magnetization directed along the present Earth's field (Fig. 10). The declination is northerly, and the inclination, although similar in magnitude to those at Cape Parkins and Victoria Lake, is directed upward (Table 2).

6 Age and origin of magnetizations

In the Cape Parkins section, the **A** and **B** magnetization directions are reversed from one another, which presumably reflects reversal of the geomagnetic field. **A** magnetization dominates. In the fine-grained dyke (site 22), **B** is almost certainly younger than **A**. Thus it appears that during the magnetization of the 1800 m section the geomagnetic field was predominantly of one polarity.

Except for the steeply dipping southern localities (sites 28, 29, 30, Table 1, Fig. 2), declinations within the thick Cape Parkins sequence are consistent, but they differ from those at inland localities. The average inclinations, after correction for panel attitudes, are all in good agreement regardless of sign (Cape Parkins 42°, Victoria Lake 41°, Holberg quarry −49°), whereas their agreement with respect to horizontal (72°, 13° and −62°) is poor. The magnetization, therefore, predates tilting. The age of tilting is not precisely known, but is probably associated with faults of Late Cretaceous to Early Tertiary age (Muller *et al.* 1974). Large differences in declination occur among the Cape Parkins, Victoria Lake and Holberg localities. They are probably too great to be caused by palaeosecular variation and may be attributed to relative rotations about local vertical axes. Victoria Lake localities have **A** magnetization (downward inclination) and Holbert a **B** magnetization (upward inclination). All magnetizations have directions that are very different from any known directions of mid-Cretaceous to Tertiary overprints in the Western Cordillera (Irving, Monger & Yole 1980). Hence **A** and **B** magnetizations are probably older than mid-Cretaceous. The minor **B** magnetizations (those with high unblocking temperature at sites, 34, 44, 45, 46P, 25) which are carried by haematite, are probably slightly younger secondary magnetizations acquired during deuteric alteration and deposition of haematite in the thickening volcanic pile at a time when the field was of opposite polarity to that when the beds were laid down. Alternatively, they could be related to the emplacement of the Island Intrusions. At most sites, however, the haematite and magnetite magnetizations are precisely parallel (Fig. 3), both presumably being acquired as a result of high-temperature auto-oxidation of the lavas during extrusion. There is no independent proof of this timing, but the ideal magnetizations depicted in Figs 3 and 4 show no evidence of later disturbance. Thus the **A** and **B** magnetizations are regarded as recorders of the palaeofield sometime in the middle Early Jurassic (Sinemurian and/or Pliensbachian).

The **C** magnetization, carried by haematite in the reddened steeply dipping beds, has distributed unblocking temperatures, and coexists with typical **A** magnetization (Figs 6 and 7). It is presumably secondary, and could have been acquired during cooling following the heating event associated with the Island Intrusions. **C** could therefore be Late Jurassic or Early Cretaceous in age. It is unlikely that **C** magnetization post-dates this event, because the inclinations are low and the majority of the tilting probably occurred after the mid-Cretaceous.

7 Tectonic interpretation; palaeolatitudinal displacement

The palaeolatitudinal displacement and rotation of Vancouver Island relative to ancestral North America are estimated by comparing the Bonanza data with determinations from

middle Lower Jurassic rock-units of the craton. Four factors influence this discussion. The first, already noted, is the recent downward revision of the geological time-scale. We shall use the time-scale of Kent & Gradstein (1986), which gives limits for the Late Triassic as 230 and 208 Ma, and those for the Early Jurassic as 208 and 187 Ma. The time-scale of Harland *et al.* (1982) places the Triassic–Jurassic boundary earlier, at 213 Ma. Previous time-scales proposed a younger age for this boundary; for example, that of van Eysinga (1975) gave 195 Ma. This change in the time-scale has the effect of diminishing displacement estimates.

The second factor is the unsatisfactory state of the cratonic palaeomagnetic record for the Early Mesozoic. Many rock-units have been studied, but much of the previous work, especially on red beds from the southwestern U.S.A., was undertaken to establish reversal stratigraphies, not primarily for tectonic studies. Emphasis was placed on 'blanket cleaning' of large numbers of samples in order to obtain wide stratigraphic coverage, rather than on detailed analysis of samples from fewer localities, the requirement for tectonic work. It is worth bearing in mind the comment by Reeve (1975, p. 257) regarding this data from red beds of the Colorado Plateau (K and W of Table 5) data that is one of the bases for the analysis of Gordon *et al.* (1984) discussed below. He noted that the peculiarities of the pole positions he obtained could have been caused by small amounts of secondary magnetization acquired prior to tilting of the beds studied and not completely removed during subsequent thermal demagnetization procedures. This is but one example of the numerous complexities of the magnetization of red beds which needs to be addressed before detailed tectonic conclusions are drawn. Furthermore data from the Colorado Plateau used for developing the cratonic record is suspect because of possible rotation of the plateau relative to the craton (see below) and, because it is derived entirely from sedimentary rocks which could be subject to inclination error.

The third factor affecting displacement estimates is the recognition, mainly as a consequence of the work of Smith & Noltimier (1979), that during the earlier part of the Jurassic, substantial apparent polar wander occurred.

The fourth factor concerns the effect that different analytical procedure have on determinations. By one procedure (designated **method I**) cratonic palaeopoles, representative for a given interval, are averaged and their mean calculated. Single intervals may be considered, or a sequence of intervals can be assembled and running means calculated to generate a cratonic apparent polar wander (apw) reference path. Running means using windows of 30 Ma duration have been given by Irving & Irving (1982) and their estimate for 190 Ma (the Early Jurassic in the time scale of van Eysinga used by them) is given in Table 4 (entry 15). Harrison & Lindh's (1982) determination using the same window is given in entry 16. The two are in excellent agreement. Hillhouse & Grommé (1984) have provided an average palaeopole for the Middle Triassic to Early Jurassic, covering about 40 Ma (entry 17, Table 4). Their palaeopole is about 5° different from the two reported above, but still within limits of error. Method I, when applied to data from intervals of rapid apparent polar wander, may introduce undesirable smoothing if the window-length chosen is too great.

By another procedure (Gordon, Cox & O'Hare 1984) palaeopoles for much longer intervals (order 100 Ma) are fitted to small circles and corresponding euler poles calculated, the assumption being that major crustal plates moved along trajectories that were constant for 100 Ma, so that palaeopoles would fall sequentially on small circles (**method II**). Their palaeopole for the Early Jurassic falls about 13° of longitude further west than palaeopoles obtained by method I (Table 4, entry 18). Their palaeopole was derived from a small circle that is calculated using data from rocks ranging in age from Early Jurassic to Late Cretaceous. Apparent polar wander paths obtained by method II are 'model' paths, being obliged by a tectonic model to lie on small circles. The procedure is questioned because

significant changes in plate trajectories occur more frequently than every 100 Ma (about every 25 Ma according to Engebretson 1982). Also it is questioned on philosophical grounds: firstly, it is not reasonable to expect that information about the Early Jurassic can be obtained from summaries that incorporate data from Cretaceous rocks; secondly, the assumption prejudges the nature of the processes that are being investigated.

In the comparison now made, method I is used and smoothing is minimized by narrowing the window as far as is possible within the accuracy of geochronological and stratigraphic dating. Tectonic models are eschewed for the reasons just given. Bonanza data will simply be compared with determinations from cratonic rocks which, present evidence indicates, were deposited during the Sinemurian and Pleinsbachian only.

Before making these comparisons, it is necessary to consider, as already mentioned, the

Table 4. Palaeopoles for Lower Jurassic peri-Atlantic tholeiites and cratonic reference palaeopoles

	Rock-unit	Location	Palaeopole (error)
(1)	Trans-Avalon dyke, Nwfd (200 Ma K–Ar)	47.1N, 53.4W	73N, 088E (4)
(2)	Shelburne dyke, N.S. (197 Ma K–Ar)	44.0N, 65.5W	69N, 098E (5)
(3)	North Mountain basalt, N.S. (200 Ma K–Ar)	45.0N, 64.7W	69N, 108E (7)
(4)	Connecticut Valley rocks (193 Ma K–Ar)	41.5N, 75.0W	65N, 087E (10)
(5)	Pre-folding intrusives CT., MA. (191 Ma K–Ar)	41.4N, 72.9W	63N, 083E (3)
(6)	Newark intrusives (200 Ma K–Ar)	40.3N, 75.9W	64N, 105E (3)
(7)	Mean of 1 to 6 (n = 6)	–	68N, 095E (5)
(8)	Intrusives stable Morocco (187–180 Ma K–Ar)	31.0N, 06.5W	65N, 116W (3) 69N, 106E*
(9)	Dykes northern Mauritania	24.0N, 08.0W	69N, 128W (4) 66N, 092E*
(10)	Dykes, southern Mauritania	16.5N, 07.5W	71, 120W (6) 69N, 088E*
(11)	Liberian dykes (192–173 Ma K–Ar)	6.7N, 10.2W	69N, 118W (5) 70N, 096E*
(12)	Mean 8 to 11 (n = 4)	–	69N, 096E (4)
(13)	Overall mean (n = 10)	–	68N, 095E (3)
(14)	Alternative mean (n = 5)	–	67N, 095E (5)
(15)	190 ± 15	–	66N, 093E (5)
(16)	190 ± 15	–	66N, 094E (4)
(17)	LTr-EJ	–	61N, 093E (5)
(18)	190 Ma	–	62N, 080E (?)

* In North American frame, obtained by rotation of −78.8° about 66.2N, 12.4W (Le Pichon, Sibuet & Francheteau 1977). (1) Hodych & Hayatsu 1980; (2) Larochelle & Wanless 1966, 197 Ma K–Ar; (3) Larochelle 1967, Carmichael & Palmer 1968; (4) de Boer & Snider 1979; (5) Smith & Noltimier 1979; (6) Opdyke 1961, Beck 1972; (8), (9) and (10) compiled by Sichler *et al.* 1980; (11) Dalrymple, Grommé & Wright 1975; (14) mean of 4, 5, 6, 8, 11 see text; (15) Early Jurassic from Irving & Irving (1982) using the time scale of van Eysinga (1975); (16) Harrison & Lindh (1982); (17) mean for Late Triassic and Early Jurassic from Hillhouse & Grommé (1984); (18) Early Jurassic from Gordon, Cox & O'Hare (1984) who used the time-scale of Harland *et al.* (1982).

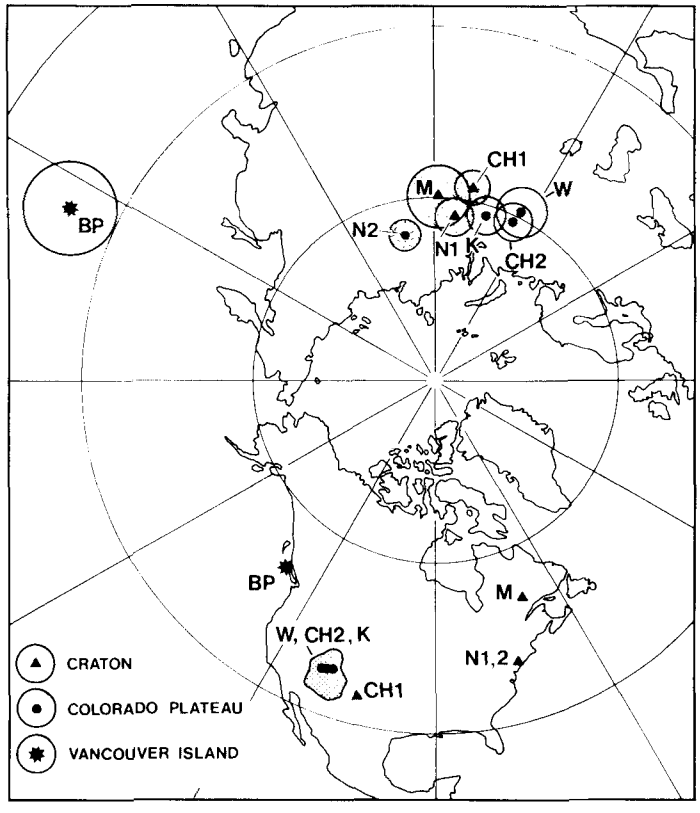


Figure 11. Palaeopoles from Upper Triassic and Lower Jurassic rocks from cratonic North America, from the Colorado Plateau (stippled) and from the Bonanza Group of the Cape Parkins section (BP). Results from the craton are as follows: CH1 Chinle Formation; M Manicouagan impact crater; N1 older Newark; N2 younger Newark. A fourth possible cratonic palaeopole, from the Popo Aige Formation (Grubbs & van der Voo 1976), is not used because it is based on only 10 samples from one site situated close to the tectonically disturbed Rocky Mountains overthrust zone. Results from the Colorado Plateau are as follows: CH2 Chinle Formation; W Wingate Formation; K Kayenta Formation. Details in text and Table 5.

Triassic and Jurassic data from the Colorado Plateau that has commonly been used to derive cratonic reference apw paths. Hamilton (1981) proposed 5–7° post mid-Cretaceous clockwise rotation on geological grounds. Steiner (1984) showed that palaeopoles from Phanerozoic rocks of various ages from the plateau are displaced by about 13° clockwise from palaeopoles from elsewhere on the craton, but she found no displacement of palaeopoles from Permian rocks. Irving & Strong (1985), however, argued that Devonian redbeds of the Colorado Plateau studied by Elston & Bressler (1977), remagnetized by the Kiaman (latest Palaeozoic), indicate a clockwise rotation of 14° ± 4. Both these estimates are a little greater than those proposed by Hamilton. Bryan & Gordon (1985) derived a smaller clockwise rotation (4° ± 3) by averaging determinations from rocks ranging from Pennsylvanian to Cretaceous in age. Their estimate is compatible with that of Hamilton. Their analysis employed data of variable quality, smoothed over 200 Ma, so effects of rotations occurring at different times are suppressed. The present concern is with early Mesozoic events, therefore, when considering the problem of rotation of the Colorado

Table 5. Other palaeopoles used in analysis

	Rock-unit	Palaepole (error)
CH1	Chinle Formation, New Mexico, Upper Triassic	58N, 079E (3)
M	Manicougan Impact Crater, 210 Ma, Quebec	60N, 089E (5)
N1	Older Newark intrusions & lavas, 195 Ma	63N, 083E (3)
N2	Younger Newark intrusions & lavas, 179 Ma	65N, 103E (3)
K	Kayenta Formation, Utah, Lower Jurassic	62N, 073E (3)
W	Wingate Formation, Utah, Lower Jurassic	59N, 063E (4)
CH2	Chinle Formation, Utah, Upper Triassic	61N, 064E (3)

Radiometric ages are given when available. CH1 Reeve & Helsley (1972); CH2 see W; M Robertson (1967), Larochelle & Currie (1967), K–Ar date 210 ± 4 Ma (Wolfe 1971) and Rb–Sr mineral isochron 210 ± 8 Ma (Jahn, Floran & Simonds 1978); N1 and N2 Smith & Noltimier (1979), K–Ar ages by Sutter & Smith (1979) and Seidemann *et al.* (1984); K Steiner & Helsley (1974); CH2 and W Reeve (1975).

Plateau, attention is confined to Late Triassic and Early Jurassic determinations only (Fig. 11, Table 5). We compare data on and off the plateau for short time intervals rather than following the smoothing technique over long intervals employed by Bryan & Gordon (1985).

There are three North American palaeopoles obtained from widely spaced localities outside of the Colorado Plateau, the older Newark rocks (N1) the rocks of the Manicouagan impact crater (M) and red sedimentary rocks of the Chinle Formation of New Mexico (CH1). There is good agreement between palaeopoles from older lavas and intrusions of the Newark Supergroup (denoted by N1 in Fig. 11, age 195 ± 5 Ma or Pliensbachian) and from the Manicouagan impact structure (M, 210 ± 8 Ma or latest Triassic to earliest Jurassic). Palaeopole (CH1) from the Chinle Formation of New Mexico (Late Triassic) is somewhat more distant from the craton than N1, as if the inclination is too shallow by 5 to 10°. As the Chinle data are from sedimentary rocks, inclination errors are always a possibility. Alternatively, the difference between CH1 and N1 could reflect apparent polar wander during the latest Triassic and Early Jurassic. However, the difference is small, and the Chinle palaeopole from New Mexico evidently belongs to the cratonic group.

Palaepoles from the Colorado Plateau (red sedimentary rocks of the Kayenta K, Chinle CH2 and Wingate W formations) lie roughly 10 to 20° to the west, near 60°N, 70°E. The differences are systematic. For the Chinle Formation, determinations (CH1 and 2) were obtained from the same rock-unit both on and off the plateau. Gordon *et al.* (1984) prefer to regard the plateau palaeopoles as a trace of apparent polar wander, but, inspection of Fig. 11 shows that there is a *prima facie* case for suggesting about 10° clockwise rotation of the plateau (Steiner 1984). Before these uncertainties (possible inclination error, incomplete removal of secondary magnetization in these sedimentary rocks, and possible rotation of the plateau) can be satisfactorily discussed, detailed restudies of these red-bed units involving the examination of sedimentary structures and demagnetization through the full range of stabilities present followed by component analysis is required (e.g. Zijderveld 1967; Roy & Park 1974). Until this work is done, it seems unwise to use palaeopoles from the Colorado Plateau to define the Early Mesozoic polar wander path for cratonic North America.

Truly cratonic data for the Sinemurian and Pliensbachian (about 205–195 Ma) are listed in Table 4 (entries 1–6) and it is these that are used for comparison with the Bonanza Group. They are from massive tholeiitic lavas and intrusives of eastern North America, mainly from the Newark Supergroup, and hence are not, like sedimentary rocks of the

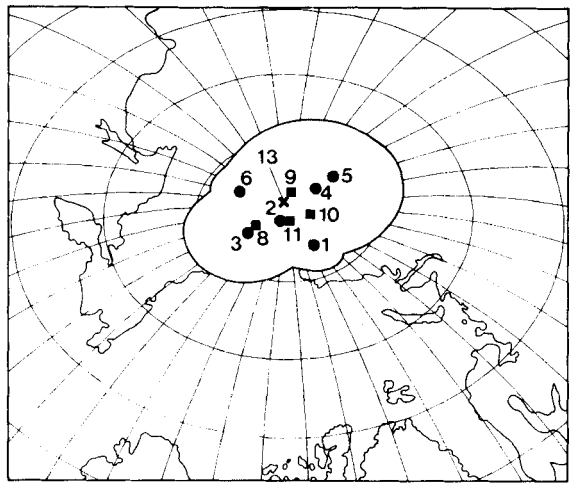


Figure 12. Palaeopoles for peri-Atlantic tholeiites and Lower Jurassic volcanics of Australia after restoration to Pangaea. Rotations used are given in caption to Fig. 13. Palaeopoles numbered as in Table 4. Pole of projection is given in entry 13 of Table 4.

Plateau, subject to inclination error. The rocks are dated accurately, both stratigraphically and by Ar^{39}_{40} studies. The determinations may be compared with coeval data from Africa rotated into the common framework of Wegener's Pangaea [Table 4 (entries 8–11), Fig. 12]. All are in good agreement. The means for the African and North American data (entries 7 and 12 of Table 4) differ by 1°. It is suggested therefore that the overall mean (entry 13) is a good reference palaeopole for the middle Early Jurassic (Sinemurian and Pliensbachian) of North America. The determination from younger Newark rocks (N2 of Smith & Noltimier 1979, Fig. 11) is not included in this average because they yield radiometric dates of about 175 Ma or Middle Jurassic, which is outside the relevant time-window.

Weaknesses in this analysis are that the dykes and basalts of entries 1 to 3 of Table 4 may not provide adequate averaging of the palaeosecular variation, and that the two results from Mauritania (entries 9 and 10) are not directly dated radiometrically (Sichler *et al.* 1980). The former are substantial bodies and the data accurate, although the time represented may be insufficient. Admission of the Mauritanian dykes into the required time-frame rests on their palaeomagnetic agreement with other African data (Sichler *et al.* 1980), not a powerful reason. However the question is not critical, because if these five determinations are excluded, the overall mean result is changed by only 1° (entry 14, Table 4).

Recently May & Butler (1986) have reviewed problems concerned with the comparison of reference path and cordilleran determinations, and have adopted a procedure (not different in principle from method I used here) by which individual cratonic palaeopoles for discrete intervals of time are compared with cordilleran palaeopoles for the same intervals. They however, applied the procedure differently. Unlike us, they accepted data from the Colorado Plateau and compared determinations from Sinemurian and Pleinsbachian rocks of the Cordillera with those from the Kayenta and Wingate formations (Table 5, Fig. 11). We have preferred to make comparisons with the Newark Supergroup and its equivalents because they belong indubitably to the craton, they have very good age control, and the data (like our own) is from massive igneous rocks and hence not subject to the possibility of inclination error as are data from sedimentary rocks of the plateau.

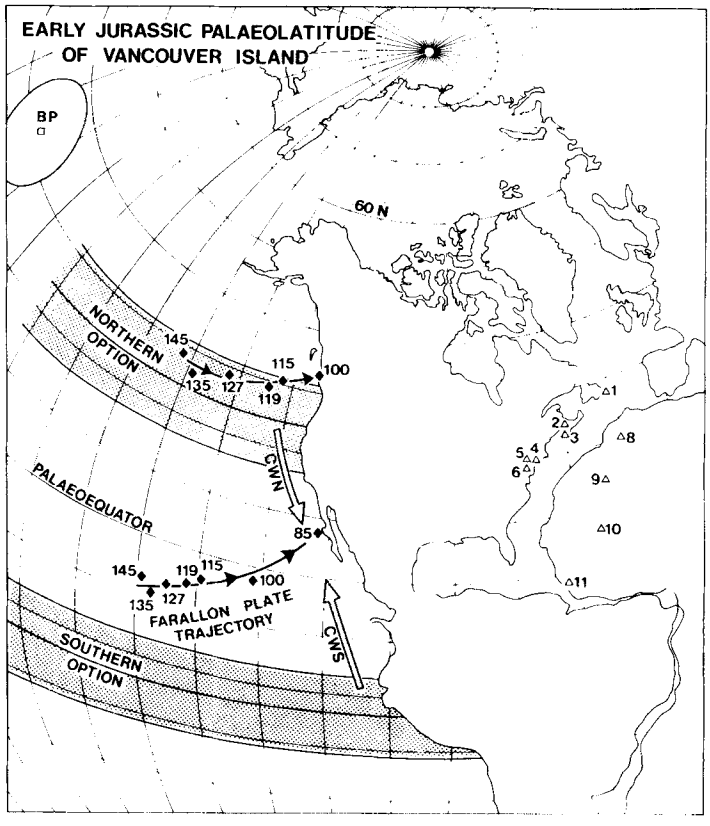


Figure 13. Tentative palaeolatitude map for the Early Jurassic. Entry 13 of Table 4 is the pole used. Rotations are as follows: Africa to North America −78.8° about 66.2°N, 12.4°W (LePichon, Sibuet & Francheateau 1977); South America to Africa 57.5° about 45.5°N, 32.2°W (Rabinowitz & LeBreque 1979); Iberia to Eurasia −23° about 50.0°N, 3.3°E (LePichon *et al.* 1971); Eurasia to North America −26.7° about 85°N, 129.9°E (LePichon *et al.* 1977); Greenland to North America −11° about 71.5°N, 126.3°E (Srivastava 1978). The northern and southern options for palaeolatitudinal belts for the Bonanza volcanics of the Cape Parkins section (BP) are stippled. The two solid arrowed lines, calibrated in Ma, are possible trajectories of Vancouver Island for the Middle Jurassic to Late Cretaceous, calculated from data of Engebretson, Cox & Gordon (1986), assuming (upper) it travelled on the Farallon plate and docked at 85 Ma in a position off Baja California, and assuming (lower) it docked at 100 Ma in its present latitude. Alternative coastwise trajectories (CWN, CWS) for the Early Jurassic to mid-Cretaceous are also shown. The sampling localities for the peri-Atlantic tholeiites are numbered as in Table 4 and Fig. 13.

8 Tectonic interpretation; rotations

Figure 11 shows that the palaeopole from the Cape Parkins section is far to the left of those from the craton, indicating a very large rotation. The distances (the north polar distances or palaeocolatitudes) between Vancouver Island and the Bonanza, and Lower Jurassic cratonic palaeopoles are comparable, but there is a small difference. The palaeolatitudinal belts for Vancouver Island are shown in Fig. 13 in which the mean palaeopole of Fig. 12 was used to draw the grid for ancestral North America. Because reversals of the palaeofield are frequent in the Early Jurassic, the data do not determine whether Vancouver Island was in the northern or southern hemisphere. If it were in the northern hemisphere (northern option of Fig. 13) the rotation of the Cape Parkins section would be 66 ± 7° anticlockwise, and the

palaeolatitudinal displacement 9 ± 7° from the south, which is significant, if our estimate of the cratonic field is correct. If Vancouver Island were in the southern hemisphere, the rotation would be 114 ± 7° clockwise, and the palaeolatitudinal displacement 57 ± 7° from the south. The notion that displacement was from the south is consistent with evidence from the distribution of Jurassic ammonite faunas of British Columbia (Tipper 1981, 1984). Taylor *et al.* (1984) and Cameron & Tipper (1985) argue that a northern, rather than a southern palaeolatitude, is the more probable, because these faunas have exclusively boreal representatives.

Geological evidence indicates that the various terranes of the western Cordillera of British Columbia (including Vancouver Island) were amalgamated together by the mid-Cretaceous (Monger, Price & Templeman-Kluit 1982). This superterrane was accreted to the margin of North America in the latest Cretaceous or Palaeocene (Irving, Monger & Yole 1980). During the mid-Cretaceous, according to palaeomagnetic data, this superterrane was situated about 2300 km south of its present position and moved northward to its present location sometime in the late Cretaceous or Early Tertiary (Beck & Noson 1972; Irving *et al.* 1985). Much of the palaeomagnetic data is derived from intrusions for which direct determinations of palaeohorizontal cannot be made, and so also can be explained by assuming a regional tilt of the Cordilleran by 30° to the southwest (Beck, Burmester & Schoonover 1981; Irving *et al.* 1985). Hence there are several possible trajectories for Vancouver Island for the interval Late Jurassic to Early Cretaceous. If translation is accepted as the correct explanation for the Cretaceous data, then the superterrane (which included Vancouver Island) would have been situated at the latitude of Baja California about 85 Ma (Fig. 13). If Vancouver Island was accreted to North America from the Farallon Plate, its trajectory can be calculated from the data of Engebretson, Cox & Gordon (1986) and is plotted in Fig. 13. Because magnetic anomalies older than mid-Jurassic cannot be confidently identified, it is not possible to extend the trajectory back as far as the Early Jurassic, so a direct comparison with Bonanza palaeolatitudes cannot be made. None the less, Vancouver Island tracks back to within 10° of the latitude belt of the southern option, by the mid-Jurassic. It is also possible that Vancouver Island was not carried on the Farallon Plate in the Jurassic and earliest Cretaceous, but on some other plate, as yet unidentified, which travelled southward along the coast of Pangaea from the northern palaeolatitude belt (CWN Fig. 13). A third possibility is that Vancouver Island originated in the southern hemisphere and travelled coastwise from the south to its Cretaceous position (CWS, Fig. 13). Finally, it is possible that the apparently divergent Cretaceous palaeopoles were produced by systematic tilt, and thus the corresponding track for the Farallon plate indicates an origin for the Wrangellian terrane from the northern palaeolatitude belt (Fig. 13). The southern trajectory has been terminated at 85 Ma, and the northern at 100 Ma to illustrate the uncertainty in 'docking' time. The hemispheric ambiguity for Vancouver Island in the late Triassic and Jurassic remains.

One matter, however, seems clear from these observations, that rotations about vertical axes have been large and variable (Fig. 14). They are comparable in magnitude to those observed in Permian, Jurassic and Triassic rocks in the Intermontane Belt (IMB of Fig. 1) of north central British Columbia (Monger & Irving 1980; Irving & Monger 1986). The magnitude of the rotations is much greater than the uncertainties in the cratonic reference field. They contrast markedly with the systematic rotations of about 60° clockwise obtained from mid-Cretaceous rocks of the western Cordillera (Irving *et al.* 1985). Therefore they pre-date the mid-Cretaceous and postdate the Early Jurassic. It is suggested that these large and variable rotations are a consequence of the processes by which the various terranes of the western Cordillera in Canada became amalagamated prior to the mid-Cretaceous. One

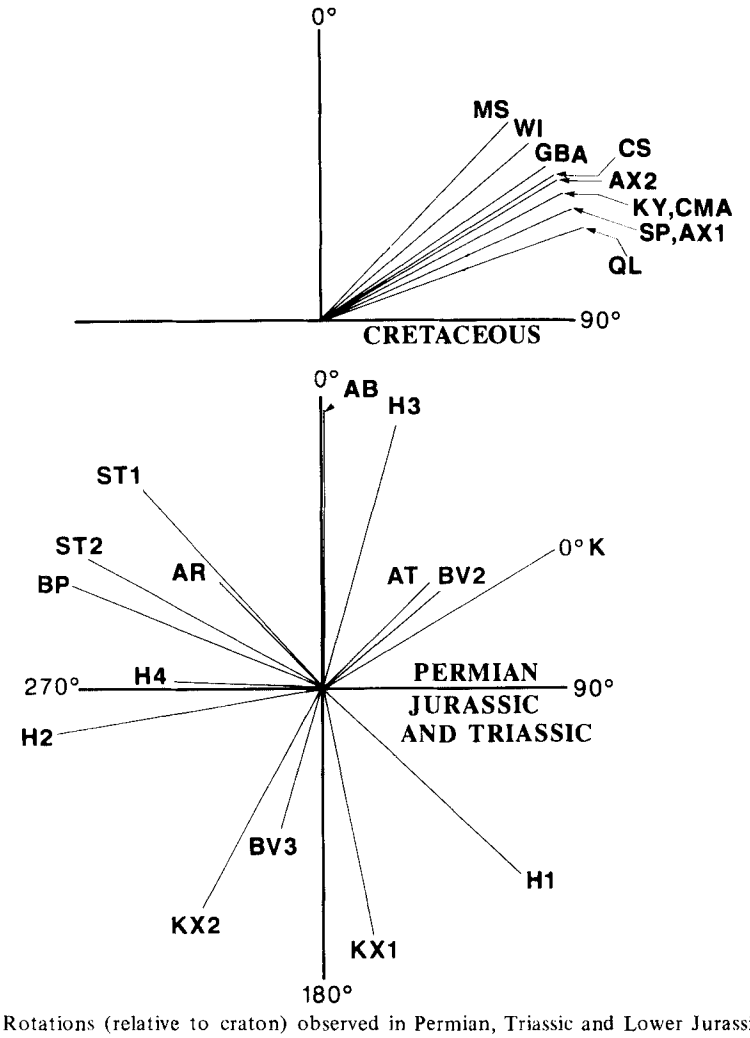


Figure 14. Rotations (relative to craton) observed in Permian, Triassic and Lower Jurassic rocks of the western Canadian Cordillera compared with those in Cretaceous rocks. All calculations are for the northern hemisphere option. Lines represent the palaeomeridian direction. If no rotation relative to the craton had occurred the lines would cluster about the zero line. Rotations are reduced to a central point in the Canadian Cordillera (Smithers 54.8°N, 127.2°W). Upper determinations are from mid-Cretaceous rock-units or from overprints presumed to be of this age in older rocks as documented in Irving *et al.* (1985). Data below are from Upper Triassic (ST Takla Group two sections; KX Karmutsen Formation two sections) and Lower Jurassic (HZ Hazelton Group three sections; B Bonanza Group, BP Cape Parkins, BV2 Holberg, BV3 Victoria Lake) and Permian (Asitka Group, three localities AB, AR, AT) calculated for the northern palaeolatitude option. Full lines are from extensive collections. Lines of half-length from fewer than 5 sites. Calculations from data of Monger & Irving (1980), Schwarz *et al.* (1980), Yole & Irving (1980), Irving & Monger (1987). Reproduced from Irving & Wynne (1987).

possibility is that during amalgamation, the various terranes were emplaced along large flat thrusts as comparatively thin, nappe-like structures between which large and variable rotations occurred. Comparable, highly variable rotations have been observed in nappes elsewhere (Channel, Catalano & D'Argenio 1980; Grubbs & van der Voo 1976). Further possibilities are that they record oroclinal bending within the nascent mobile belt, and that they record internal deformation after amalgamation.

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